Software Documentation

1. Software Overview

- Primary Function: The software acts as a real-time signal processor. It continuously polls the MAX30102 sensor for infrared reflectance data, filters that data to identify the human pulse, and triggers synchronized auditory (PWM) and visual (NeoPixel) feedback loops.

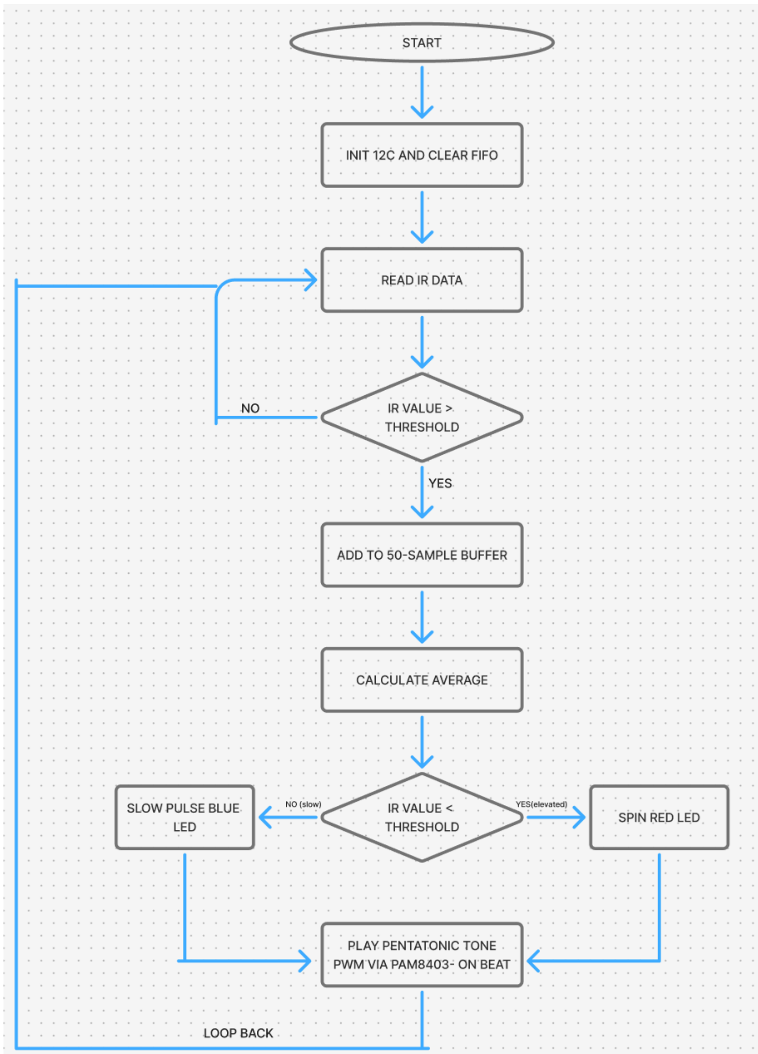
  Platforms Used:

- MicroPython (v1.27): Used for core logic due to its efficiency in handling I2C registers and hardware interrupts on the ESP32.
- Thonny IDE: Used for real-time debugging, script uploading, and monitoring the serial plotter to visualize the pulse waveform.
- Blender: Used for the digital fabrication of the enclosure. We designed a custom record player style box specifically to house the 12V-to-5V power conversion components and provides a steady mount for the sensor and NeoPixel ring.

2. System Logic

1. Hardware Stabilization: Upon boot, the script explicitly sets the speaker and NeoPixel pins to a "Low" state to prevent the electrical "pop" or "hum" caused by the high-current Buck Converter powering up.
2. I2C Handshake: The ESP32 initializes a SoftI2C bus. A critical step here is the Register Reset where the script sends a specific command (0x40) to the sensor to clear its memory. This prevents the KCKC serial noise caused by leftover data from previous runs.
3. Data Acquisition: The script enters a while True loop, reading 3 bytes of data from the sensor's FIFO (First-In, First-Out) buffer.
4. Signal Filtering (Rolling Average): Because raw sensor data is "noisy" (affected by how hard the user presses or ambient light), the software stores the last 50 readings in a list. It calculates the average of these 50 points to create a "baseline."
5. Beat Detection: A "beat" is triggered only when a new IR reading drops significantly below the baseline (indicating blood has surged into the finger).
6. Action Trigger: * Visual: Sends a color-fill command to the NeoPixel ring.
   - Audio: Calculates the time since the last beat to determine BPM, then maps that BPM to a specific frequency in a Pentatonic Scale (e.g., C4, D4, E4, G4, A4).
7. Auto-Kill: If the IR value drops below a certain threshold (finger removed), the script immediately clears the NeoPixels and silences the PWM to prevent "ghost" tones.

3. Flowchart



(NOT AI)(Made on Figma)

4. App Flow (If MIT App Inventor)

N/A